MESA COHORT C4

AI and its Applications — Section 1 Final Examination

PROBLEM STATEMENT BOOKLET

Instructions to Students

1. Read your assigned problem statement carefully.
2. Your solution must use live external data wherever possible. The system must work with some data at the time of submission.
3. All AI tasks must use Gemini Flash. Do not use Claude API.
4. Deploy your solution on Vercel. Submit a working public URL along with your GitHub repository link and the Components Map.

Medium Difficulty

YouTube Trend Radar

Creator Economy • Content Strategy

Ishaan has been running a tech YouTube channel for eighteen months and posts three videos per week. His content covers AI tools, productivity apps, and tech reviews for Indian audiences. His best-performing videos aren't always the ones he thought would work — they're the ones that caught a trending wave at the right time. A video on Cursor IDE he dropped when that tool went viral got 40,000 views in three days. A video on an app he reviewed two months after its peak got 600 views. Timing the content to the trend cycle matters more than he initially realised.

Every Sunday evening before he plans his week's filming, Ishaan spends ninety minutes manually browsing YouTube's trending pages and searching for recent uploads in his content categories. It works to a degree, but it's pattern-blind. He can see individual videos that are doing well, but he can't easily see the meta-pattern across fifty trending videos at once — which topic clusters are

surfacing this week, which formats are winning, how long the videos in a given trend tend to run.

What Ishaan actually needs is pattern-level intelligence, not a list of videos. He wants a Sunday morning brief that tells him which topics are clustering in his content space this week — which subjects multiple creators are covering simultaneously, which formats seem to be performing well right now, what the audience is actively engaging with. Three to five such theme clusters, each with enough context to inform a filming decision, is more useful to him than ninety minutes of manual browsing. His content categories and preferred region should be configurable, so the brief stays relevant as his channel evolves. If this arrives before Sunday noon, he can walk into Monday's filming session knowing exactly what to cover and why this week is the right moment.

> Build and deploy an AI-powered Sunday content strategy brief for Ishaan, delivered each week before he begins filming.